



Bed Demand & Capacity Analysis

Albion Care Network · January 2024 – December 2025

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5 hospitals · 8 wards · 130,993 admissions

[View the live Power BI dashboard](#)



PROJECT OVERVIEW

About Albion Care Network

- **Leading UK private healthcare provider**
39+ hospitals delivering elective and emergency care across complex, multi-ward environments.
- **Highly variable bed demand**
Each specialty has distinct admission patterns and length-of-stay profiles, making capacity management a central operational challenge.
- **Digital infrastructure exists but falls short**
EHRs, ADT systems, and bed management tools support reporting — not forward-looking planning.
- **Forecasting is the missing piece**
Transforming operational data into predictive intelligence so hospitals anticipate constraints rather than react to them. Bed availability directly impacts patient safety, elective cancellations, and revenue.

The Problem We Were Asked to Solve

Five specific obstacles driving the need for a predictive, data-driven approach to bed capacity planning.

Reactive Bed Management

Capacity decisions are often made only after bed shortages occur, limiting the ability to prevent disruption.

Fragmented Data Systems

Admissions, discharge, bed management, and staffing data exist in separate systems, preventing a unified, forward-looking view of capacity.

High Variability in Demand

Emergency and elective admissions fluctuate daily and seasonally, making static planning approaches ineffective.

Limited Predictive Insights

Operational planning relies heavily on historical averages rather than forecasts that anticipate future demand.

Operational Risk Across Units

ICU, general wards, and outpatient units experience mismatches between demand and available capacity, increasing safety and efficiency risks.



The Bottom Line

Five headline numbers from the Executive Dashboard

74.6%

Average Occupancy Rate

191

Beds Available

33.6%

Hours At Capacity

80%

Emergency Admission %

4.3%

Safe Ratio Breach Rate

Demand Peaks Every Winter

Occupancy rate by month, 2024–2025 — a seasonal swing the forecast also confirms



81.98%

Winter Occupancy

65.19%

Summer Occupancy

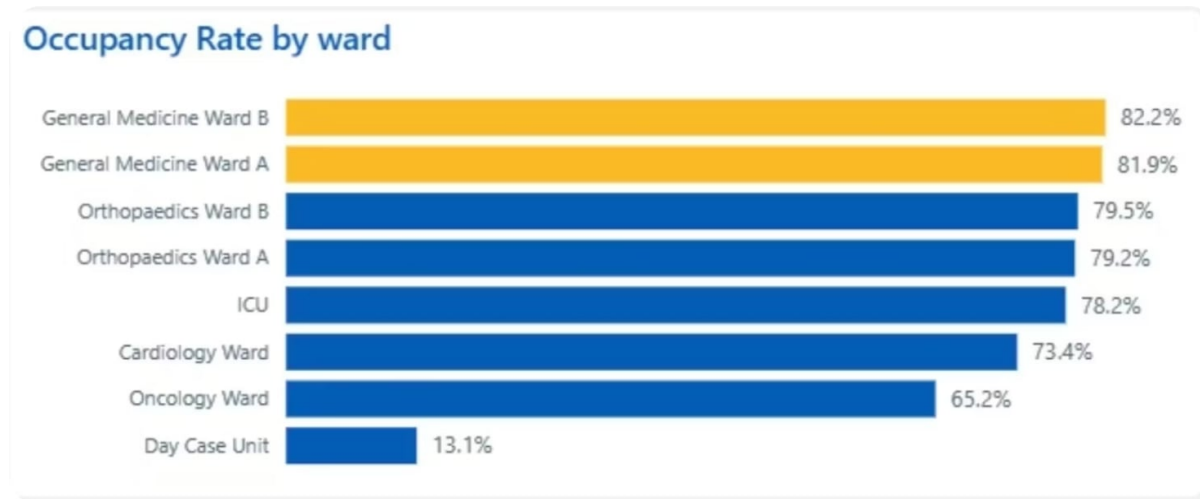
16.79pp

Seasonal Swing

January peaks at 38,676 ED arrivals — August drops to 23,432.
The curve repeats every year.

The Average Hides the Real Pressure

Occupancy by ward — General Medicine wards run hottest, with both above 80%



General Medicine Ward B (82.2%) and Ward A (81.9%) are the hottest wards in the network — both approaching the 85% safe limit.

Only Day Case Unit has meaningful spare capacity, Day Case shows 13% — but that's by design, not spare capacity; those patients go home the same day.

82.2%

General Medicine Ward B

81.9%

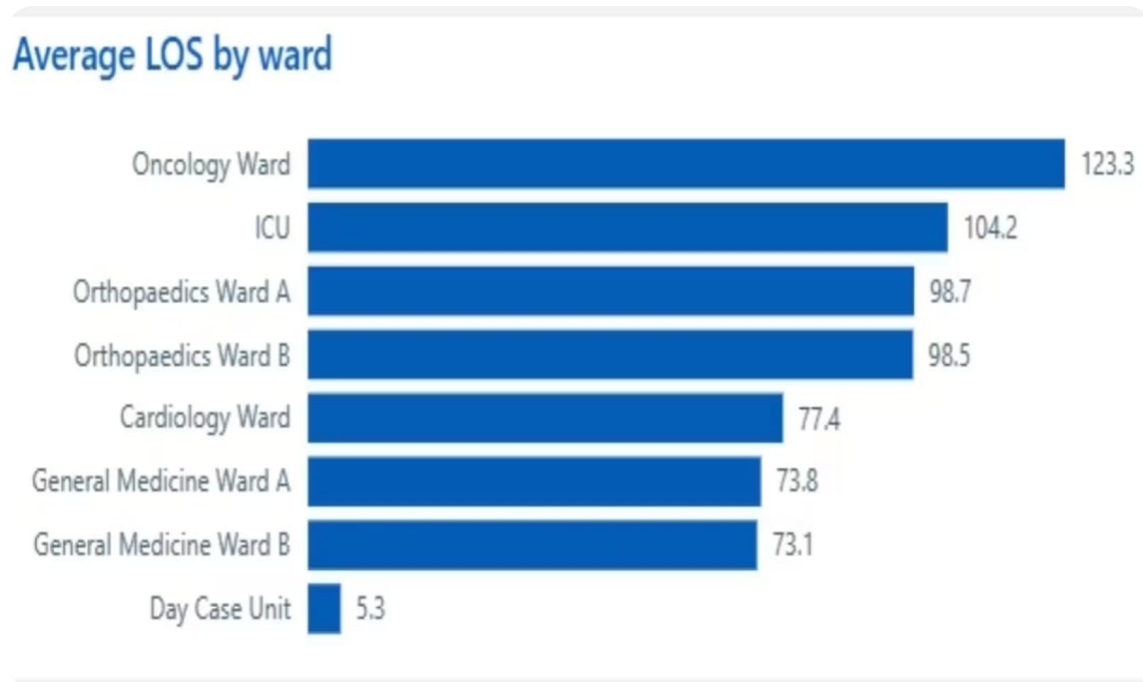
General Medicine Ward A

13.1%

Day Case Unit

Why These Wards Stay Full

Average Length of Stay by ward (hours)



123.1

Oncology LOS (hours)

5.3

Day Case LOS (hours)

Long-stay wards block flow. Day Case Unit has room to flex.

Staffing Stays Flat While Demand Swings

The single biggest operational risk — a fixed workforce facing a variable workload

Planned vs Actual Staff



4.3%

Safe Ratio Breach

Network average

16.79pp

Seasonal Swing

Summer to winter

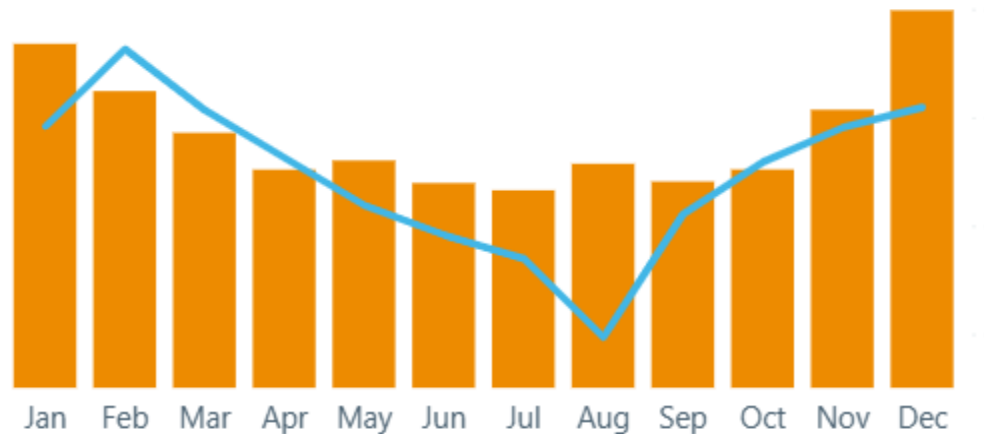
Staffing is planned as a flat annual number while demand swings 17pp from summer to winter. Safe-ratio breaches nearly double in winter — the highest-risk period.

Elective Cancellations Track Occupancy

Cancellations rise and fall with ward occupancy — winter is the danger zone

Elective cancellations vs occupancy

● Elective Cancel Rate ● Occupancy Rate



Peak: December–January

Cancellations peak at nearly 12% in December when occupancy is at 80.9%. Unplanned demand crowds out scheduled care.

Fix: shift electives to May–Aug

Occupancy drops to 59–68% in summer. This is the window to protect planned care.

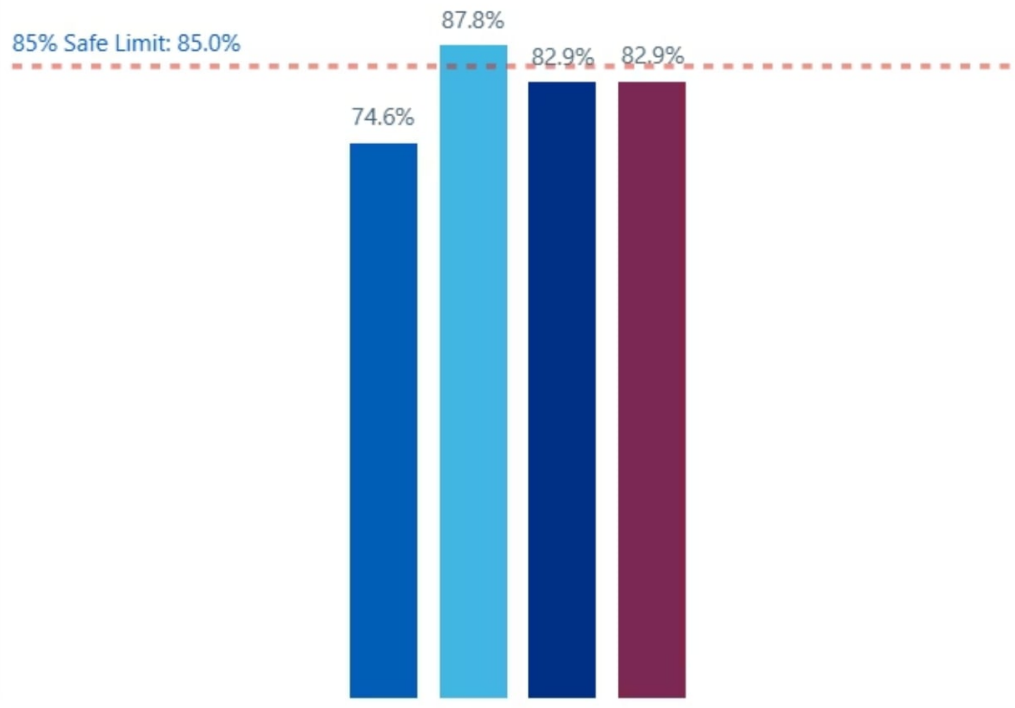
We Can See the Impact Before It Happens

Scenario planning — occupancy under four scenarios

Occupancy under each scenario — red = past safe limits

Occupancy under each scenario

● Occupancy Rate ● Flu Surge ● Delayed Discharge ● Bed Closure



A moderate flu surge alone tips the network past its safe limit — and winter brings all three pressures at once.

87.8%

+20% Emergency Surge

a single moderate surge alone pushes us *past* the safe limit

82.9%

Discharge crisis (+10%)

bed *flow* is a risk too; delays fill beds without a single extra patient

Winter Crisis - Each pressure on its own already pushes us past the safe limit — and in a real winter, all three happen together, so the network runs out of room

ICU, Orthopedics and General Medicine hit capacity first in every scenario



Recommendations

Three evidence-based interventions — each directly tied to the data

1. Staffing Adjustments

Align cover to the seasonal demand curve

2. Bed-Usage Optimization

Accelerate discharge on long-stay wards

3. Elective Schedule Management

Reschedule into quieter months

Each recommendation is grounded in the EDA findings and can be actioned using the live Power BI dashboard already in place.



Rec 1 — Staffing Adjustments

Align staffing to the seasonal demand curve

The problem

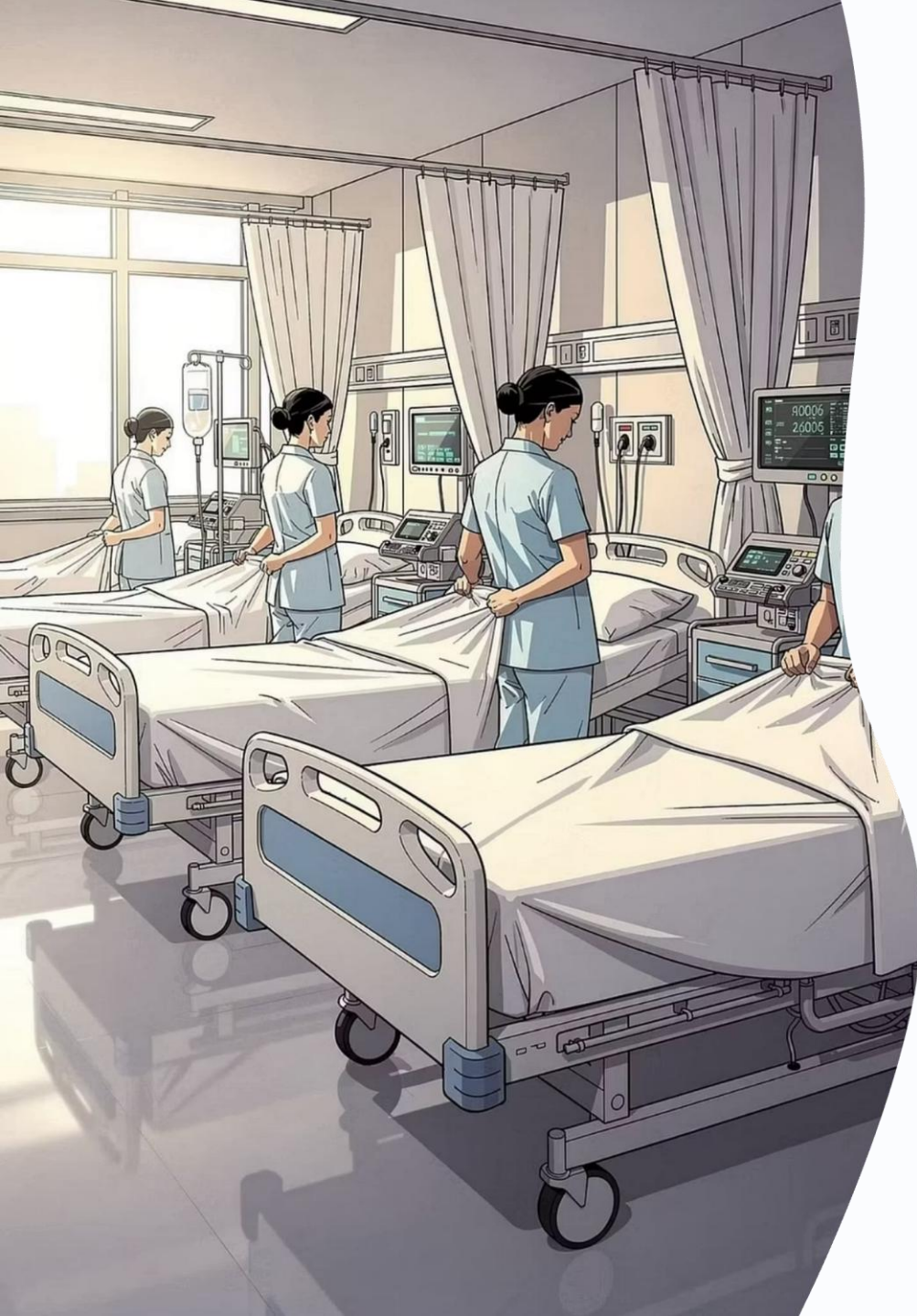
Staffing is flat year-round while demand swings 16.79pp — safe-ratio breaches reach 4.3%, nearly double in winter.

The action

Add winter cover to General Medicine and ICU for Dec–Feb, triggered at 85% occupancy.

The target

Reduce safe-ratio breaches from 4.3% to below 2% by winter 2026.



Rec 2 — Bed-Usage Optimisation

Accelerate discharge on long-stay wards

The problem

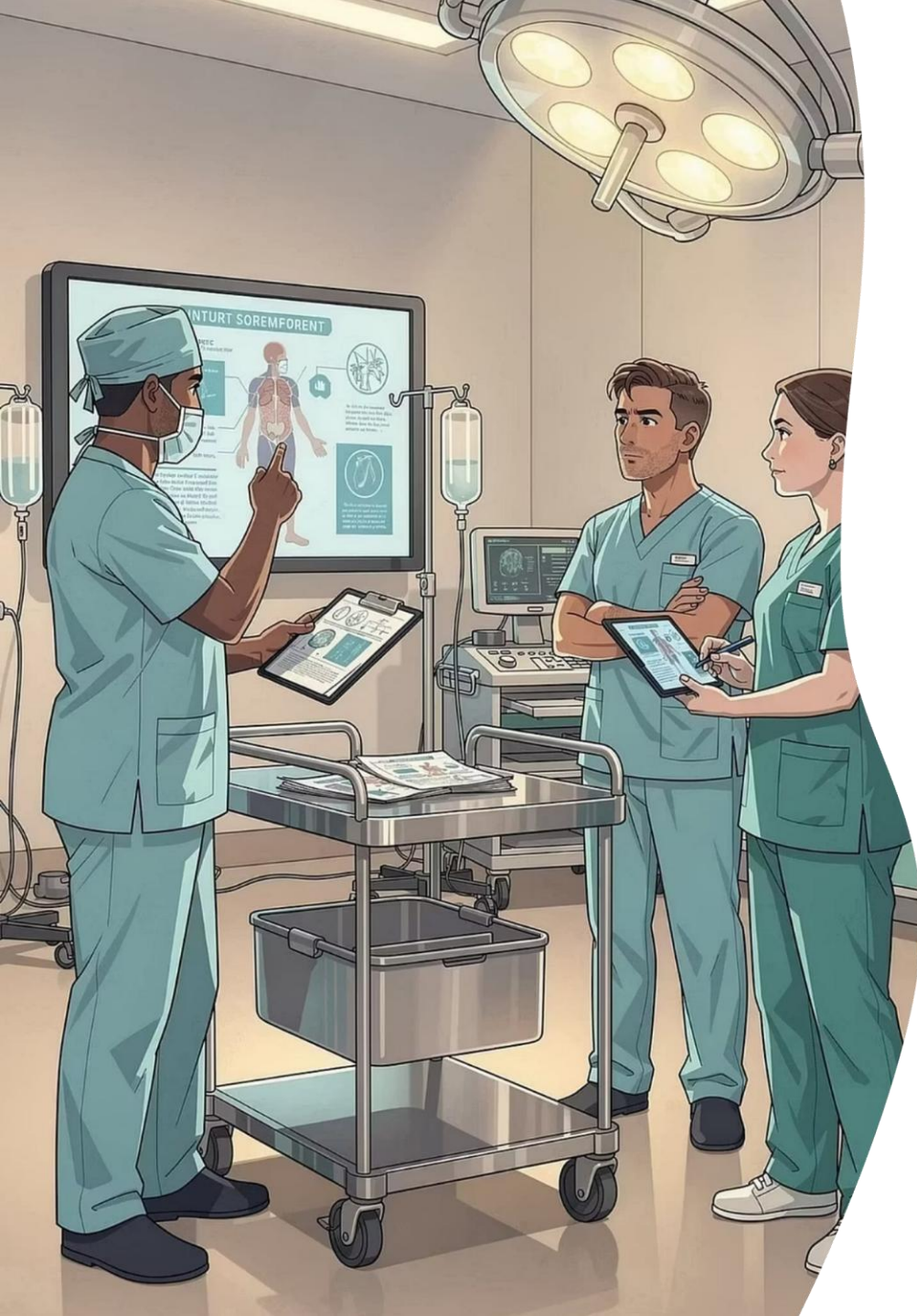
Oncology (123.1 hours), ICU (104.2 hours) and Orthopaedics (~98 hours) hold beds longest — blocking flow across the network.

The action

Introduce daily discharge planning rounds on these wards and repurpose the Day Case Unit as a winter flex buffer.

The target

Reduce Oncology LOS by 10% and raise Day Case utilisation to 40%+ in Dec-Feb.



Rec 3 — Elective Schedule Management

Reschedule into quieter months — don't cancel

The problem

Cancellations spike in Dec–Jan when occupancy peaks, wasting theatre time and delaying patient care.

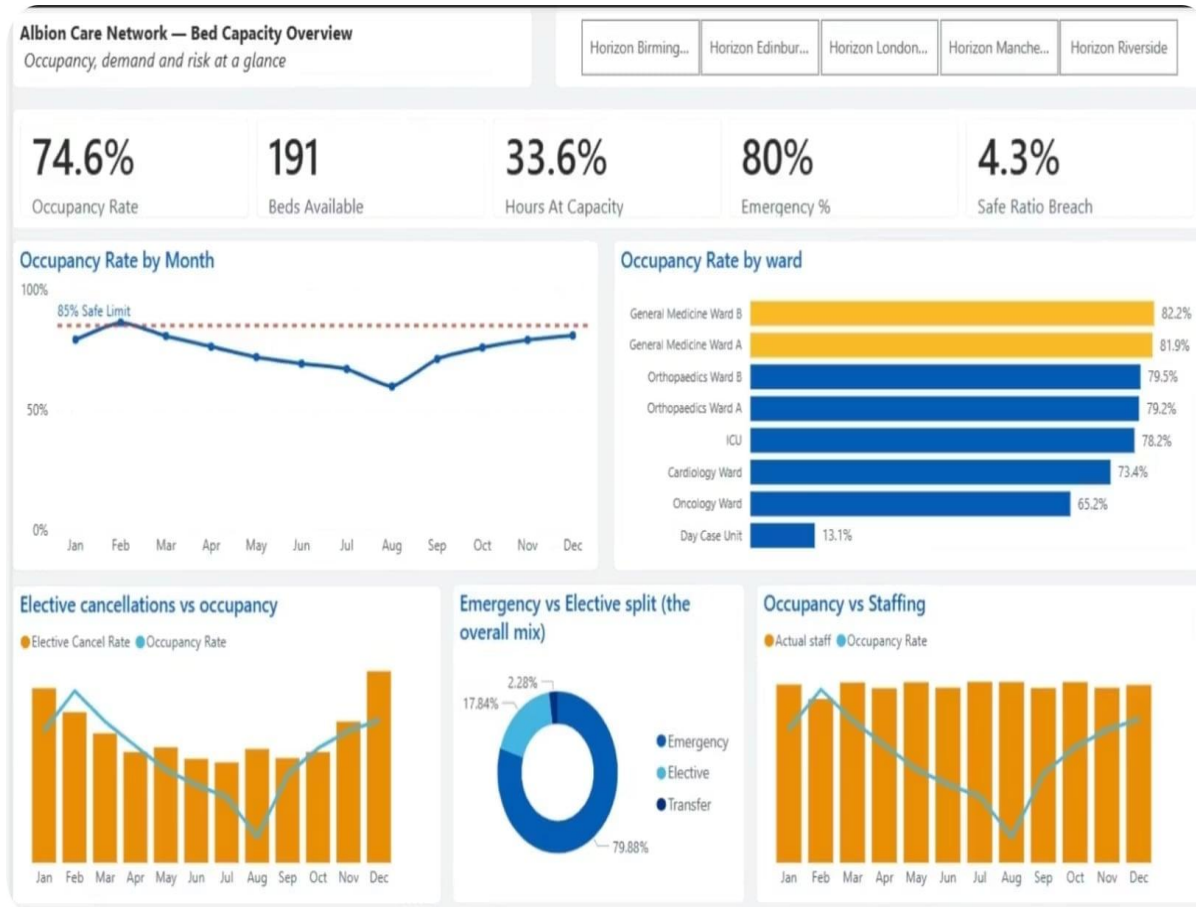
The action

Move electives into May–September (occupancy 59–68%) and book against forecast demand.

The target

Reduce winter cancellation rate by 30% and increase summer elective throughput by winter 2026.

One Dashboard, Always Up to Date



EDAs

Occupancy trends, demand drivers, cancellations & ED arrivals



Executive & Operational

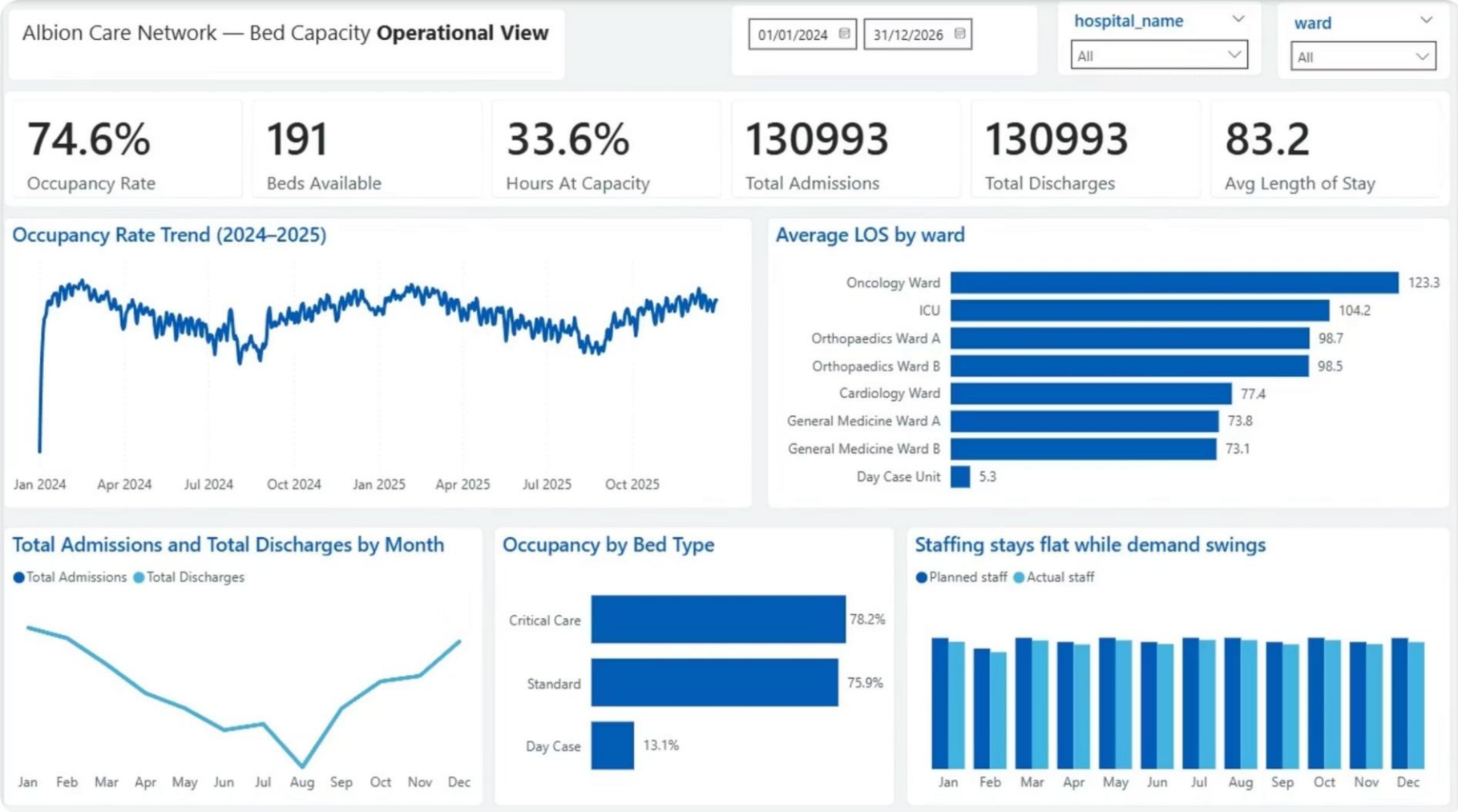
Live KPIs: 74.6% occupancy, 191 beds, 4.3% breach rate



Scenario Planning

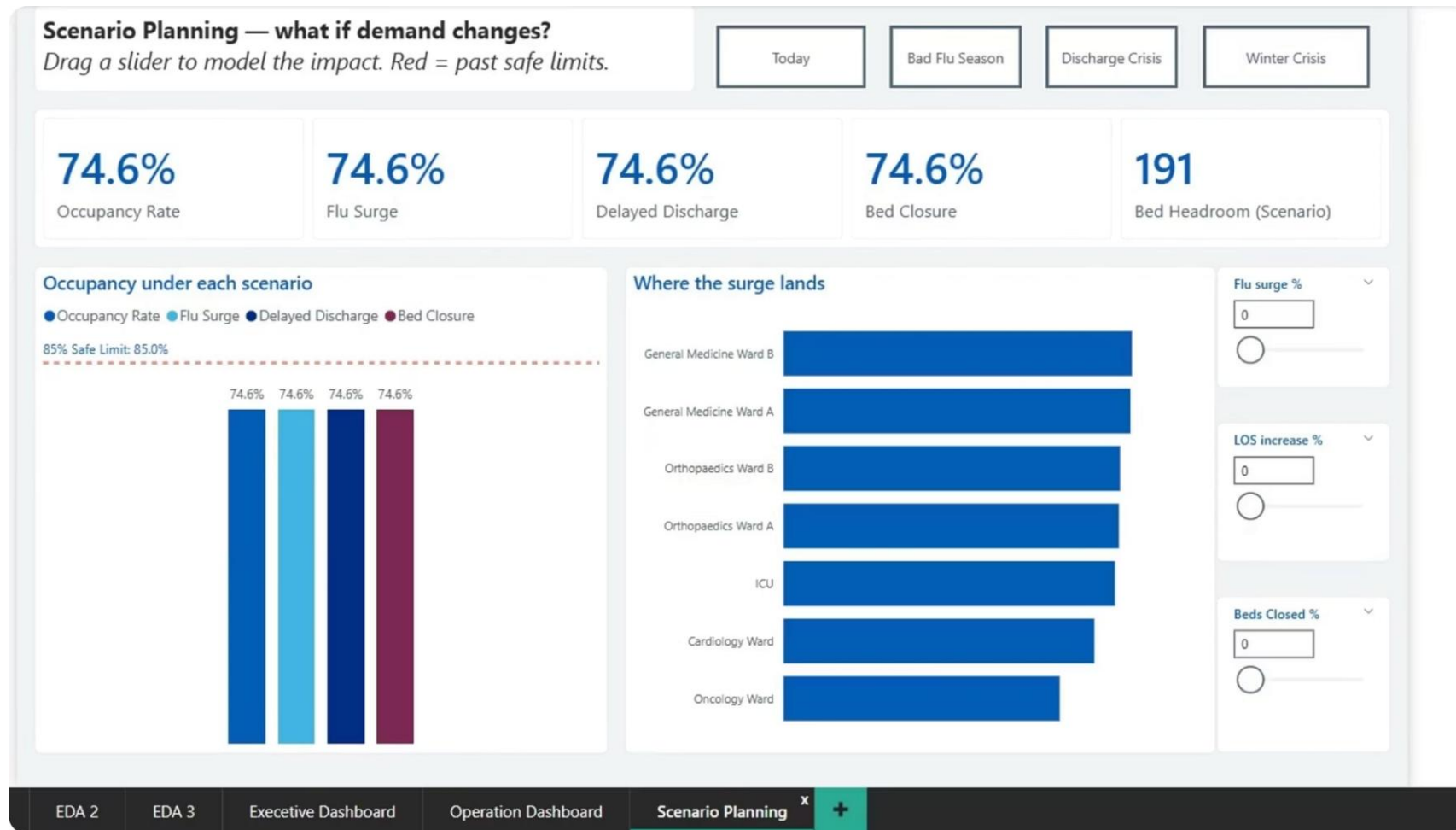
Flu surge, discharge crisis & winter crisis simulations

Albion Care Network — Bed Capacity Operational View



Scenario Planning — What if demand changes?

Drag a slider to model the impact. Red = past safe limits.





What Happens Next

Turning insight into everyday practice

Use the dashboard weekly

Monitor occupancy against the 85% trigger every week from October onwards.

Pilot the three recommendations

Winter staffing uplift, discharge planning on long-stay wards, elective rescheduling into summer.

Deploy the XGBoost forecast

Short-term (24–72h) and medium-term (7–30 day) bed demand forecasts per ward unit.

Set up continuous monitoring

Predicted-vs-actual tracking with quarterly model retraining.



Thank You

We appreciate your time and attention.
Together, we can build a smarter, more resilient hospital
system.

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Questions? Let's continue the conversation.